Adrian M. Price-Whelan — Publication List

Current position: Associate Research Scientist
Center for Computational Astrophysics, Flatiron Institute,
162 Fifth Ave., New York, NY 10010, USA
adrianmpw@gmail.com adrian.pw github.com/adrn arXiv

Publications — *ADS search*

Refereed: 165 articles (18 first author)
Citations 59820
h-index: 60
(as of *2026-10-26*)

Refereed

165 Tavangar, K. *et al.* (incl. **APW**), *Phase Spirals across Galactic Disks. II. Using Large-scale "Macrospirals" in Phase-spiral Amplitude to Derive Perturbation Times*, ApJ, 1010, 126, 2026

164 Arora, A. *et al.* (incl. **APW**), *No Stream Left Unscathed: The Imprint of a Host Galaxy*, ApJ, 1008, 91, 2026 (arXiv:2605.16200) [4 citations]

163 Holm, J. K. *et al.* (incl. **APW**), *Evidence for the first globular cluster stellar stream beyond the Milky Way*, Nature, 656, 843, 2026 (arXiv:2608.12254)

162 Rose, J. C. *et al.* (incl. **APW**), *The DREAMS Project: Disentangling the Impact of Halo-to-halo Variance and Baryonic Feedback on Milky Way Satellite Galaxies*, ApJ, 1006, 244, 2026 (arXiv:2512.02095) [3 citations]

161 Brooks, R. A. N. *et al.* (incl. **APW**), *The Milky Way–Large Magellanic Cloud interaction with simulation-based inference*, MNRAS, 550, 2026 (arXiv:2510.04735) [9 citations]

160 Brooks, R. A. N. *et al.* (incl. **APW**), *Quantifying the Milky Way, LMC and their interaction using all-sky kinematics of outer halo stars*, MNRAS, 549, 2026 (arXiv:2603.08788) [5 citations]

159 SDSS Collaboration *et al.* (incl. **APW**), *The Nineteenth Data Release of the Sloan Digital Sky Survey*, ApJS, 285, 9, 2026 (arXiv:2507.07093) [96 citations]

158 Sagear, S.; Ballard, S.; Daniel, K. J.; **Price-Whelan, A. M.** *et al.*, *The Orbital Eccentricities of Planets in the Kinematic Thin and Thick Galactic Disks*, AJ, 172, 42, 2026

157 Nibauer, J.; Bonaca, A.; **Price-Whelan, A. M.**; Spergel, D. N. *et al.*, *Measurement of Substructure from the Kinematics of the GD-1 Stellar Stream*, ApJ, 1004, 62, 2026 (arXiv:2510.02247) [23 citations]

156 Chen, Y.; Gnedin, O. Y.; **Price-Whelan, A. M.**, *StarStream on Gaia: Stream Discovery and Mass-loss Rate of Globular Clusters*, ApJS, 283, 60, 2026 (arXiv:2510.14924) [9 citations]

155 Lucey, M.; Mateu, C.; **Price-Whelan, A. M.**; Hogg, D. W. *et al.*, *Inferring Stellar Densities with Flexible Models. I. The Distribution of RR Lyrae in the Milky Way with Gaia DR3*, AJ, 171, 249, 2026 (arXiv:2510.03221) [4 citations]

154 Horta, D.; **Price-Whelan, A. M.**; Koposov, S. E.; Hunt, J. A. S. *et al.*, *The Milky Way's Circular Velocity Curve Measured Using Element Abundance Gradients*, ApJ, 1000, 8, 2026

(arXiv:2601.18876) [4 citations]

153 Hughes, J. *et al.* (incl. **APW**), *The Milky Way Bulge Extratidal Star Survey: NGC 6569*, AJ, 171, 137, 2026 (arXiv:2512.19074) [2 citations]

152 Otto, J. M. *et al.* (incl. **APW**), *The Open Cluster Chemical Abundances and Mapping Survey. VIII. Galactic Chemical Gradient and Azimuthal Analysis from SDSS/MWM DR19*, AJ, 171, 91, 2026 (arXiv:2507.07264) [29 citations]

151 Brooks, R. A. N. *et al.* (incl. **APW**), *A simulation-based inference approach to the dynamics of the MW–LMC system - validation*, MNRAS, 545, 2026 (arXiv:2507.10667) [16 citations]

150 Kollmeier, J. A. *et al.* (incl. **APW**), *Sloan Digital Sky Survey. V. Pioneering Panoptic Spectroscopy*, AJ, 171, 52, 2026 (arXiv:2507.06989) [137 citations]

149 Hunt, J. A. S. *et al.* (incl. **APW**), *The dark matter wake of a galactic bar revealed by multichannel singular spectral analysis*, MNRAS, 545, 2026 (arXiv:2510.09751) [6 citations]

148 Chen, Y.; Gnedin, O. Y.; **Price-Whelan, A. M.**; Holm-Hansen, C., *StarStream: Automatic Detection Algorithm for Stellar Streams*, ApJ, 995, 15, 2025 (arXiv:2510.14929) [6 citations]

147 Viswanathan, A.; Horta, D.; **Price-Whelan, A. M.**; Starkenburg, E., *A slow spin to win: The gradual kinematic evolution across metallicities of the proto-Galaxy to the high-α disc*, A&A, 703, 2025 (arXiv:2411.12165) [14 citations]

146 Oeur, M.; Loebman, S. R.; **Price-Whelan, A. M.**; Arora, A. *et al.*, *OTI on FIRE: Testing the Efficacy of Orbital Torus Imaging to Recover the Galactic Potential*, ApJ, 993, 29, 2025 (arXiv:2505.05590) [2 citations]

145 Bernet, M.; Ramos, P.; Antoja, T.; **Price-Whelan, A. M.** *et al.*, *Dynamics of tidal spiral arms: Machine learning-assisted identification of equations and application to the Milky Way*, A&A, 702, 2025 (arXiv:2506.17383) [6 citations]

144 Chance, Q. *et al.* (incl. **APW**), *paired: A Statistical Framework for Detecting Stellar Binarity with Gaia RVs. I. Sensitivity to Unresolved Binaries*, ApJ, 992, 131, 2025 (arXiv:2206.11275) [12 citations]

143 Adams, D. K. *et al.* (incl. **APW**), *Semianalytic Modeling of Dark Matter Subhalo Encounters with Thin Stellar Streams: Statistical Predictions for GD-1-like Streams in Cold Dark Matter*, ApJ, 991, 66, 2025 (arXiv:2412.13144) [8 citations]

142 Sheffield, A. A. *et al.* (incl. **APW**), *Tracing Red Giant Members of the Globular Cluster Palomar 5 with APOGEE and Gaia*, ApJ, 991, 30, 2025 (arXiv:2507.21212)

141 Wagg, T. *et al.* (incl. **APW**), *Delayed and Displaced: The Impact of Binary Interactions on Core-collapse SN Feedback*, AJ, 170, 192, 2025 (arXiv:2504.17903) [13 citations]

140 Anguiano, B. *et al.* (incl. **APW**), *Tidal Debris Candidates from the ω Centauri Accretion Event and Its Role Building Up the Milky Way Halo*, ApJ, 989, 12, 2025 (arXiv:2505.08353) [4 citations]

139 Filion, C. *et al.* (incl. **APW**), *Counterculture Stars: Slow and Retrograde Stars with Low-alpha Disk Abundances*, ApJ, 989, 70, 2025 (arXiv:2506.09927)

138 Tavangar, K.; **Price-Whelan, A. M.**, *Inferring the Density and Membership of Stellar Streams with Flexible Models: The GD-1 Stream in Gaia Data Release 3*, ApJ, 988, 45, 2025 (arXiv:2502.13236) [13 citations]

137 Frankel, N.; Hogg, D. W.; Tremaine, S.; **Price-Whelan, A. M.** *et al.*, *Iron Snails: Nonequilibrium Dynamics and Spiral Abundance Patterns*, ApJ, 987, 81, 2025 (arXiv:2407.07149) [19 citations]

136 Horta, D.; **Price-Whelan, A. M.**; Hogg, D. W.; Ness, M. K. *et al.*, *Lux: A Generative, Multi-output, Latent-variable Model for Astronomical Data with Noisy Labels*, AJ, 169, 314, 2025 (arXiv:2502.01745) [10 citations]

135 Bonaca, A.; **Price-Whelan, A. M.**, *Stellar streams in the Gaia era*, New Astronomy Reviews, 100, 101713, 2025 (arXiv:2405.19410) [103 citations]

134 Griffith, E. J. *et al.* (incl. **APW**), *Many Elements Matter: Detailed Abundance Patterns Reveal Star Formation and Enrichment Differences among Milky Way Structural Components*, AJ, 169, 280, 2025 (arXiv:2410.22121) [17 citations]

133 Widrow, L. M. *et al.* (incl. **APW**), *An Equilibrium Model of the Galaxy Determined by Element Abundance Gradients*, ApJ, 983, 134, 2025

132 Nibauer, J.; Bonaca, A.; Spergel, D. N.; **Price-Whelan, A. M.** *et al.*, *StreamSculptor: Hamiltonian Perturbation Theory for Stellar Streams in Flexible Potentials with Differentiable Simulations*, ApJ, 983, 68, 2025 (arXiv:2410.21174) [20 citations]

131 Starkman, N.; **Price-Whelan, A. M.**; Nibauer, J., *unxt: A Python package for unit-aware computing with JAX*, JOSS, 10, 7771, 2025 (arXiv:2603.08770) [2 citations]

130 Lu, Y. *et al.* (incl. **APW**), *Evidence of Truly Young High-α Dwarf Stars*, AJ, 169, 168, 2025 (arXiv:2410.02962) [11 citations]

129 Starkman, N. *et al.* (incl. **APW**), *Stream Members Only: Data-driven Characterization of Stellar Streams with Mixture Density Networks*, ApJ, 980, 253, 2025 (arXiv:2311.16960) [13 citations]

128 **Price-Whelan, A. M.**; Hunt, J. A. S.; Horta, D.; Oeur, M. *et al.*, *Data-driven Dynamics with Orbital Torus Imaging: A Flexible Model of the Vertical Phase Space of the Galaxy*, ApJ, 979, 115, 2025 (arXiv:2401.07903) [9 citations]

127 Foote, H. R. *et al.* (incl. **APW**), *Segue 2 Recently Collided with the Cetus-Palca Stream: New Opportunities to Constrain Dark Matter in an Ultra-faint Dwarf*, ApJ, 979, 171, 2025 (arXiv:2408.06415) [4 citations]

126 Wagg, T.; Breivik, K.; Renzo, M.; **Price-Whelan, A. M.**, *cogsworth: A Gala of COSMIC proportions combining binary stellar evolution and galactic dynamics*, JOSS, 10, 7400, 2025 [6 citations]

125 Wagg, T.; Breivik, K.; Renzo, M.; **Price-Whelan, A. M.**, *cogsworth: A Gala of COSMIC Proportions Combining Binary Stellar Evolution and Galactic Dynamics*, ApJS, 276, 16, 2025 (arXiv:2409.04543) [14 citations]

124 Brooks, R. A. N.; Garavito-Camargo, N.; Johnston, K. V.; **Price-Whelan, A. M.** *et al.*, *LMC Calls, Milky Way Halo Answers: Disentangling the Effects of the MW–LMC Interaction on Stellar Stream Populations*, ApJ, 978, 79, 2025 (arXiv:2410.02574) [21 citations]

123 Hackshaw, Z. *et al.* (incl. **APW**), *[X/Fe] Marks the Spot: Mapping Chemical Azimuthal Variations in the Galactic Disk with APOGEE*, ApJ, 977, 143, 2024 (arXiv:2405.18120) [27 citations]

122 Sagear, S.; **Price-Whelan, A. M.**; Ballard, S.; Lu, Y. (. *et al.*, *zoomies: A Tool to Infer Stellar Age from Vertical Action in Gaia Data*, ApJ, 977, 49, 2024 (arXiv:2403.09878) [7 citations]

121 Garavito-Camargo, N.; **Price-Whelan, A. M.**; Samuel, J.; Cunningham, E. C. *et al.*, *On the Corotation of Milky Way Satellites: LMC-mass Satellites Induce Apparent Motions in Outer Halo Tracers*, ApJ, 975, 100, 2024 (arXiv:2311.11359) [20 citations]

120 Foster, S. *et al.* (incl. **APW**), *Carbon enrichment in APOGEE disk stars as evidence of mass transfer in binaries*, A&A, 689, 2024 (arXiv:2407.18130) [6 citations]

119 Horta, D. *et al.* (incl. **APW**), *Stellar Mergers or Truly Young? Intermediate-age Stars on Highly Radial Orbits in the Milky Way's Stellar Halo*, ApJ, 971, 170, 2024 (arXiv:2403.09777) [19 citations]

118 Castro-Ginard, A. *et al.* (incl. **APW**), *Gaia DR3 detectability of unresolved binary systems*, A&A, 688, 2024 (arXiv:2404.14127) [112 citations]

117 Gaia Collaboration *et al.* (incl. **APW**), *Discovery of a dormant 33 solar-mass black hole in pre-release Gaia astrometry*, A&A, 686, 2024 (arXiv:2404.10486) [184 citations]

116 Cunningham, E. C.; Hunt, J. A. S.; **Price-Whelan, A. M.**; Johnston, K. V. *et al.*, *Chemical Cartography of the Sagittarius Stream with Gaia*, ApJ, 963, 95, 2024 (arXiv:2307.08730) [34 citations]

115 Hunt, J. A. S.; **Price-Whelan, A. M.**; Johnston, K. V.; McClure, R. L. *et al.*, *Radial phase spirals in the Solar neighbourhood*, MNRAS, 527, 11393, 2024 (arXiv:2401.08748) [17 citations]

114 Horta, D.; **Price-Whelan, A. M.**; Hogg, D. W.; Johnston, K. V. *et al.*, *Orbital Torus Imaging: Acceleration, Density, and Dark Matter in the Galactic Disk Measured with Element Abundance Gradients*, ApJ, 962, 165, 2024 (arXiv:2312.07664) [24 citations]

113 Aganze, C. *et al.* (incl. **APW**), *Prospects for Detecting Gaps in Globular Cluster Stellar Streams in External Galaxies with the Nancy Grace Roman Space Telescope*, ApJ, 962, 151, 2024 (arXiv:2305.12045) [15 citations]

112 Darragh-Ford, E.; Hunt, J. A. S.; **Price-Whelan, A. M.**; Johnston, K. V., *ESCARGOT: Mapping Vertical Phase Spiral Characteristics Throughout the Real and Simulated Milky Way*, ApJ, 955, 74, 2023 (arXiv:2302.09086) [36 citations]

111 Yavetz, T. D.; Johnston, K. V.; Pearson, S.; **Price-Whelan, A. M.** *et al.*, *Stream Fanning and Bifurcations: Observable Signatures of Resonances in Stellar Stream Morphology*, ApJ, 954, 215, 2023 (arXiv:2212.11006) [19 citations]

110 Castro-Ginard, A. *et al.* (incl. **APW**), *Estimating the selection function of Gaia DR3 subsamples*, A&A, 677, 2023 (arXiv:2303.17738) [46 citations]

109 Lucey, M. *et al.* (incl. **APW**), *Carbon-enhanced metal-poor star candidates from BP/RP spectra in Gaia DR3*, MNRAS, 523, 4049, 2023 (arXiv:2206.08299) [36 citations]

108 Silva, É. *et al.* (incl. **APW**), *Extratidal Members of Segue 3 are Rare and Difficult to Confirm*, RNAAS, 7, 127, 2023

107 Hawkins, K.; **Price-Whelan, A. M.**; Sheffield, A. A.; Subrahimovic, A. Z. *et al.*, *On the Hunt for the Origins of the Orphan-Chenab Stream: Detailed Element Abundances with APOGEE and Gaia*, ApJ, 948, 123, 2023 (arXiv:2205.14218) [15 citations]

106 Lucey, M. *et al.* (incl. **APW**), *Dynamically constraining the length of the Milky way bar*, MNRAS,

520, 4779, 2023 (arXiv:2206.01798) [52 citations]

105 Cantat-Gaudin, T. *et al.* (incl. **APW**), *An empirical model of the Gaia DR3 selection function*, A&A, 669, 2023 (arXiv:2208.09335) [93 citations]

104 Chamberlain, K.; **Price-Whelan, A. M.**; Besla, G.; Cunningham, E. C. *et al.*, *Implications of the Milky Way Travel Velocity for Dynamical Mass Estimates of the Local Group*, ApJ, 942, 18, 2023 (arXiv:2204.07173) [35 citations]

103 Rix, H.; Chandra, V.; Andrae, R.; **Price-Whelan, A. M.** *et al.*, *The Poor Old Heart of the Milky Way*, ApJ, 941, 45, 2022 (arXiv:2209.02722) [143 citations]

102 Pearson, S.; **Price-Whelan, A. M.**; Hogg, D. W.; Seth, A. C. *et al.*, *Mapping Dark Matter with Extragalactic Stellar Streams: The Case of Centaurus A*, ApJ, 941, 19, 2022 (arXiv:2205.12277) [41 citations]

101 Hunt, J. A. S.; **Price-Whelan, A. M.**; Johnston, K. V.; Darragh-Ford, E., *Multiple phase spirals suggest multiple origins in Gaia DR3*, MNRAS, 516, 2022 (arXiv:2206.06125) [69 citations]

100 Dillamore, A. M.; Belokurov, V.; Evans, N. W.; **Price-Whelan, A. M.**, *The impact of a massive Sagittarius dSph on GD-1-like streams*, MNRAS, 516, 1685, 2022 (arXiv:2205.13547) [33 citations]

99 Anguiano, B. *et al.* (incl. **APW**), *White Dwarf Binaries across the H-R Diagram*, AJ, 164, 126, 2022 (arXiv:2207.13992) [19 citations]

98 Yoshida, S.; Grunblatt, S.; **Price-Whelan, A. M.**, *Constraining the Planet Occurrence Rate around Halo Stars of Potentially Extragalactic Origin*, AJ, 164, 119, 2022 (arXiv:2206.13556) [6 citations]

97 Myers, N. *et al.* (incl. **APW**), *The Open Cluster Chemical Abundances and Mapping Survey. VI. Galactic Chemical Gradient Analysis from APOGEE DR17*, AJ, 164, 85, 2022 (arXiv:2206.13650) [82 citations]

96 Astropy Collaboration; **Price-Whelan, A. M.**; Lim, P. L.; Earl, N. *et al.*, *The Astropy Project: Sustaining and Growing a Community-oriented Open-source Project and the Latest Major Release (v5.0) of the Core Package*, ApJ, 935, 167, 2022 (arXiv:2206.14220) [5981 citations]

95 Angus, R.; **Price-Whelan, A. M.**; Zinn, J. C.; Bedell, M. *et al.*, *The 3D Galactocentric Velocities of Kepler Stars: Marginalizing Over Missing Radial Velocities*, AJ, 164, 25, 2022 (arXiv:2205.08901) [5 citations]

94 El-Badry, K. *et al.* (incl. **APW**), *Unicorns and giraffes in the binary zoo: stripped giants with subgiant companions*, MNRAS, 512, 5620, 2022 (arXiv:2203.06348) [72 citations]

93 Abdurro'uf *et al.* (incl. **APW**), *The Seventeenth Data Release of the Sloan Digital Sky Surveys: Complete Release of MaNGA, MaStar, and APOGEE-2 Data*, ApJS, 259, 35, 2022 (arXiv:2112.02026) [1496 citations]

92 Gandhi, S. S.; Johnston, K. V.; Hunt, J. A. S.; **Price-Whelan, A. M.** *et al.*, *Snails across Scales: Local and Global Phase-mixing Structures as Probes of the Past and Future Milky Way*, ApJ, 928, 80, 2022 (arXiv:2107.03562) [29 citations]

91 Eilers, A. *et al.* (incl. **APW**), *Stellar Abundance Maps of the Milky Way Disk*, ApJ, 928, 23, 2022 (arXiv:2112.03295) [57 citations]

90 Hedges, C. *et al.* (incl. **APW**), *Erratum: "TOI-2076 and TOI-1807: Two Young, Comoving Plan-*

etary Systems within 50 pc Identified by TESS that are Ideal Candidates for Further Follow Up" (2021, AJ, 162, 54), AJ, 163, 143, 2022 [3 citations]

89 Ness, M. K. *et al.* (incl. **APW**), *The Homogeneity of the Star-forming Environment of the Milky Way Disk over Time*, ApJ, 926, 144, 2022 (arXiv:2109.05722) [52 citations]

88 Hedges, C. *et al.* (incl. **APW**), *Erratum: "TOI-2076 and TOI-1807: Two young, Comoving Planetary Systems within 50 pc Identified by TESS that are Ideal Candidates for Further Follow Up" (2021, AJ, 162, 54)*, AJ, 162, 305, 2021

87 Hasselquist, S. *et al.* (incl. **APW**), *APOGEE Chemical Abundance Patterns of the Massive Milky Way Satellites*, ApJ, 923, 172, 2021 (arXiv:2109.05130) [175 citations]

86 Garavito-Camargo, N.; Patel, E.; Besla, G.; **Price-Whelan, A. M.** *et al.*, *The Clustering of Orbital Poles Induced by the LMC: Hints for the Origin of Planes of Satellites*, ApJ, 923, 140, 2021 (arXiv:2108.07321) [40 citations]

85 Santana, F. A. *et al.* (incl. **APW**), *Final Targeting Strategy for the SDSS-IV APOGEE-2S Survey*, AJ, 162, 303, 2021 (arXiv:2108.11908) [113 citations]

84 Beaton, R. L. *et al.* (incl. **APW**), *Final Targeting Strategy for the Sloan Digital Sky Survey IV Apache Point Observatory Galactic Evolution Experiment 2 North Survey*, AJ, 162, 302, 2021 (arXiv:2108.11907) [114 citations]

83 Kounkel, M.; Covey, K. R.; Stassun, K. G.; **Price-Whelan, A. M.** *et al.*, *Double-lined Spectroscopic Binaries in the APOGEE DR16 and DR17 Data*, AJ, 162, 184, 2021 (arXiv:2107.10860) [86 citations]

82 Garavito-Camargo, N.; Besla, G.; Laporte, C. F. P.; **Price-Whelan, A. M.** *et al.*, *Quantifying the Impact of the Large Magellanic Cloud on the Structure of the Milky Way's Dark Matter Halo Using Basis Function Expansions*, ApJ, 919, 109, 2021 (arXiv:2010.00816) [139 citations]

81 Rix, H. *et al.* (incl. **APW**), *Selection Functions in Astronomical Data Modeling, with the Space Density of White Dwarfs as a Worked Example*, AJ, 162, 142, 2021 (arXiv:2106.07653) [44 citations]

80 Grunblatt, S. K.; Zinn, J. C.; **Price-Whelan, A. M.**; Angus, R. *et al.*, *Age-dating Red Giant Stars Associated with Galactic Disk and Halo Substructures*, ApJ, 916, 88, 2021 (arXiv:2105.10505) [37 citations]

79 Hedges, C. *et al.* (incl. **APW**), *TOI-2076 and TOI-1807: Two Young, Comoving Planetary Systems within 50 pc Identified by TESS that are Ideal Candidates for Further Follow Up*, AJ, 162, 54, 2021 (arXiv:2111.01311) [51 citations]

78 Foreman-Mackey, D. *et al.* (incl. **APW**), *exoplanet: Gradient-based probabilistic inference for exoplanet data & other astronomical time series*, JOSS, 6, 3285, 2021 (arXiv:2105.01994) [235 citations]

77 Putman, M. E.; Zheng, Y.; **Price-Whelan, A. M.**; Grcevich, J. *et al.*, *The Gas Content and Stripping of Local Group Dwarf Galaxies*, ApJ, 913, 53, 2021 (arXiv:2101.07809) [148 citations]

76 Sheffield, A. A. *et al.* (incl. **APW**), *Chemodynamically Characterizing the Jhelum Stellar Stream with APOGEE-2*, ApJ, 913, 39, 2021 (arXiv:2103.07488) [6 citations]

75 Valluri, M.; **Price-Whelan, A. M.**; Snyder, S. J., *Detecting the Figure Rotation of Dark Matter Halos with Tidal Streams*, ApJ, 910, 150, 2021 (arXiv:2009.09004) [21 citations]

74 Miller, A. *et al.* (incl. **APW**), *Orbital and Stellar Parameters for 2M06464003+0109157: A Double-lined Eclipsing Binary of Spotted, Sub-solar Twins*, PASP, 133, 44201, 2021 (arXiv:2103.10488) [4 citations]

73 **Price-Whelan, A. M.**; Hogg, D. W.; Johnston, K. V.; Ness, M. K. *et al.*, *Orbital Torus Imaging: Using Element Abundances to Map Orbits and Mass in the Milky Way*, ApJ, 910, 17, 2021 (arXiv:2012.00015) [31 citations]

72 Yavetz, T. D.; Johnston, K. V.; Pearson, S.; **Price-Whelan, A. M.** *et al.*, *Separatrix divergence of stellar streams in galactic potentials*, MNRAS, 501, 1791, 2021 (arXiv:2011.11919) [28 citations]

71 Mazzola, C. N. *et al.* (incl. **APW**), *The close binary fraction as a function of stellar parameters in APOGEE: a strong anticorrelation with α abundances*, MNRAS, 499, 1607, 2020 (arXiv:2007.09059) [60 citations]

70 Shipp, N.; **Price-Whelan, A. M.**; Tavangar, K.; Mateu, C. *et al.*, *Discovery of Extended Tidal Tails around the Globular Cluster Palomar 13*, AJ, 160, 244, 2020 (arXiv:2006.12501) [28 citations]

69 Angus, R.; Beane, A.; **Price-Whelan, A. M.**; Newton, E. *et al.*, *Exploring the Evolution of Stellar Rotation Using Galactic Kinematics*, AJ, 160, 90, 2020 (arXiv:2005.09387) [54 citations]

68 Ahumada, R. *et al.* (incl. **APW**), *The 16th Data Release of the Sloan Digital Sky Surveys: First Release from the APOGEE-2 Southern Survey and Full Release of eBOSS Spectra*, ApJS, 249, 3, 2020 (arXiv:1912.02905) [1517 citations]

67 Caldwell, N.; Bonaca, A.; **Price-Whelan, A. M.**; Sesar, B. *et al.*, *A Larger Extent for the Ophiuchus Stream*, AJ, 159, 287, 2020 (arXiv:2004.14350) [15 citations]

66 **Price-Whelan, A. M.**; Hogg, D. W.; Rix, H.; Beaton, R. L. *et al.*, *Close Binary Companions to APOGEE DR16 Stars: 20,000 Binary-star Systems Across the Color-Magnitude Diagram*, ApJ, 895, 2, 2020 (arXiv:2002.00014) [132 citations]

65 Mumford, S. *et al.* (incl. **APW**), *SunPy: A Python package for Solar Physics*, JOSS, 5, 1832, 2020 [49 citations]

64 Hayes, C. R. *et al.* (incl. **APW**), *Metallicity and α-Element Abundance Gradients along the Sagittarius Stream as Seen by APOGEE*, ApJ, 889, 63, 2020 (arXiv:1912.06707) [85 citations]

63 Bonaca, A.; Pearson, S.; **Price-Whelan, A. M.**; Dey, A. *et al.*, *Variations in the Width, Density, and Direction of the Palomar 5 Tidal Tails*, ApJ, 889, 70, 2020 (arXiv:1910.00592) [77 citations]

62 Nidever, D. L.; **Price-Whelan, A. M.**; Choi, Y.; Beaton, R. L. *et al.*, *Spectroscopy of the Young Stellar Association Price-Whelan 1: Origin in the Magellanic Leading Arm and Constraints on the Milky Way Hot Halo*, ApJ, 887, 115, 2019 (arXiv:1910.05360) [24 citations]

61 **Price-Whelan, A. M.**; Nidever, D. L.; Choi, Y.; Schlafly, E. F. *et al.*, *Discovery of a Disrupting Open Cluster Far into the Milky Way Halo: A Recent Star Formation Event in the Leading Arm of the Magellanic Stream?*, ApJ, 887, 19, 2019 (arXiv:1811.05991) [38 citations]

60 **Price-Whelan, A. M.**; Mateu, C.; Iorio, G.; Pearson, S. *et al.*, *Kinematics of the Palomar 5 Stellar Stream from RR Lyrae Stars*, AJ, 158, 223, 2019 (arXiv:1910.00595) [44 citations]

59 Chakrabarti, S.; Chang, P.; **Price-Whelan, A. M.**; Read, J. *et al.*, *Antlia 2's Role in Driving the*

Ripples in the Outer Gas Disk of the Galaxy, ApJ, 886, 67, 2019 (arXiv:1906.04203) [21 citations]

58 Koppelman, H. H.; Helmi, A.; Massari, D.; **Price-Whelan, A. M.** *et al.*, *Multiple retrograde substructures in the Galactic halo: A shattered view of Galactic history*, A&A, 631, 2019 (arXiv:1909.08924) [264 citations]

57 Kollmeier, J. *et al.* (incl. **APW**), *SDSS-V Pioneering Panoptic Spectroscopy*, BAAS, 51, 274, 2019 [22 citations]

56 Norman, D. *et al.* (incl. **APW**), *The Growing Importance of a Tech Savvy Astronomy and Astrophysics Workforce*, BAAS, 51, 18, 2019 (arXiv:1910.08376) [2 citations]

55 Tollerud, E.; Smith, A.; **Price-Whelan, A. M.**; Cruz, K. *et al.*, *Sustaining Community-Driven Software for Astronomy in the 2020s*, BAAS, 51, 180, 2019

54 Bauer, A. *et al.* (incl. **APW**), *A Need for Dedicated Outreach Expertise and Online Programming*, BAAS, 51, 130, 2019 (arXiv:1910.14088)

53 Smith, A. *et al.* (incl. **APW**), *Elevating the Role of Software as a Product of the Research Enterprise*, BAAS, 51, 52, 2019 (arXiv:1907.06981)

52 Casey, A. R. *et al.* (incl. **APW**), *Tidal Interactions between Binary Stars Can Drive Lithium Production in Low-mass Red Giants*, ApJ, 880, 125, 2019 (arXiv:1902.04102) [103 citations]

51 Erkal, D. *et al.* (incl. **APW**), *The total mass of the Large Magellanic Cloud from its perturbation on the Orphan stream*, MNRAS, 487, 2685, 2019 (arXiv:1812.08192) [384 citations]

50 Bonaca, A.; Hogg, D. W.; **Price-Whelan, A. M.**; Conroy, C., *The Spur and the Gap in GD-1: Dynamical Evidence for a Dark Substructure in the Milky Way Halo*, ApJ, 880, 38, 2019 (arXiv:1811.03631) [260 citations]

49 Rasskazov, A. *et al.* (incl. **APW**), *Hypervelocity Stars from a Supermassive Black Hole-Intermediate-mass Black Hole Binary*, ApJ, 878, 17, 2019 (arXiv:1810.12354) [37 citations]

48 Koposov, S. E. *et al.* (incl. **APW**), *Piercing the Milky Way: an all-sky view of the Orphan Stream*, MNRAS, 485, 4726, 2019 (arXiv:1812.08172) [144 citations]

47 Williams, B. *et al.* (incl. **APW**), *Far Reaching Science with Resolved Stellar Populations in the 2020s*, BAAS, 51, 301, 2019

46 **Price-Whelan, A. M.**; Breivik, K.; D'Orazio, D.; Hogg, D. W. *et al.*, *Stellar multiplicity: an interdisciplinary nexus*, BAAS, 51, 206, 2019 [3 citations]

45 Sanderson, R. *et al.* (incl. **APW**), *The Multidimensional Milky Way*, BAAS, 51, 347, 2019 (arXiv:1903.07641) [8 citations]

44 Ness, M. *et al.* (incl. **APW**), *In Pursuit of Galactic Archaeology*, BAAS, 51, 238, 2019

43 Li, T. *et al.* (incl. **APW**), *Dark Matter Physics with Wide Field Spectroscopic Surveys*, BAAS, 51, 252, 2019

42 **Price-Whelan, A. M.**; Goodman, J., *Binary Companions of Evolved Stars in APOGEE DR14: Orbital Circularization*, ApJ, 867, 5, 2018 (arXiv:1804.06841) [36 citations]

41 De Rosa, G. *et al.* (incl. **APW**), *Velocity-resolved Reverberation Mapping of Five Bright Seyfert 1 Galaxies*, ApJ, 866, 133, 2018 (arXiv:1807.04784) [104 citations]

40 Kado-Fong, E.; Greene, J. E.; Hendel, D.; **Price-Whelan, A. M.** *et al.*, *Tidal Features at 0.05 < z < 0.45 in the Hyper Suprime-Cam Subaru Strategic Program: Properties and Formation*

Channels, ApJ, 866, 103, 2018 (arXiv:1805.05970) [72 citations]

39 Anderson, L.; Hogg, D. W.; Leistedt, B.; **Price-Whelan, A. M.** *et al.*, *Improving Gaia Parallax Precision with a Data-driven Model of Stars*, AJ, 156, 145, 2018 (arXiv:1706.05055) [31 citations]

38 Astropy Collaboration; **Price-Whelan, A. M.**; Sipócz, B. M.; Günther, H. M. *et al.*, *The Astropy Project: Building an Open-science Project and Status of the v2.0 Core Package*, AJ, 156, 123, 2018 (arXiv:1801.02634) [11156 citations]

37 Hendel, D. *et al.* (incl. **APW**), *SMHASH: anatomy of the Orphan Stream using RR Lyrae stars*, MNRAS, 479, 570, 2018 (arXiv:1711.04663) [22 citations]

36 **Price-Whelan, A. M.**; Hogg, D. W.; Rix, H.; De Lee, N. *et al.*, *Binary Companions of Evolved Stars in APOGEE DR14: Search Method and Catalog of ∼5000 Companions*, AJ, 156, 18, 2018 (arXiv:1804.04662) [1337 citations]

35 Bergemann, M. *et al.* (incl. **APW**), *Two chemically similar stellar overdensities on opposite sides of the plane of the Galactic disk*, Nature, 555, 334, 2018 (arXiv:1803.00563) [66 citations]

34 Morris, B. M. *et al.* (incl. **APW**), *astroplan: An Open Source Observation Planning Package in Python*, AJ, 155, 128, 2018 (arXiv:1712.09631) [85 citations]

33 Oh, S.; **Price-Whelan, A. M.**; Brewer, J. M.; Hogg, D. W. *et al.*, *Kronos and Krios: Evidence for Accretion of a Massive, Rocky Planetary System in a Comoving Pair of Solar-type Stars*, ApJ, 854, 138, 2018 (arXiv:1709.05344) [100 citations]

32 Sheffield, A. A.; **Price-Whelan, A. M.**; Tzanidakis, A.; Johnston, K. V. *et al.*, *A Disk Origin for the Monoceros Ring and A13 Stellar Overdensities*, ApJ, 854, 47, 2018 (arXiv:1801.01171) [38 citations]

31 Greco, J. P.; Greene, J. E.; **Price-Whelan, A. M.**; Leauthaud, A. *et al.*, *Sumo Puff: Tidal debris or disturbed ultra-diffuse galaxy?*, PASJ, 70, 2018 (arXiv:1704.06681) [22 citations]

30 Goulding, A. D. *et al.* (incl. **APW**), *Galaxy interactions trigger rapid black hole growth: An unprecedented view from the Hyper Suprime-Cam survey*, PASJ, 70, 2018 (arXiv:1706.07436) [207 citations]

29 **Price-Whelan, A. M.**, *Gala: A Python package for galactic dynamics*, JOSS, 2, 388, 2017 [369 citations]

28 Alam, S. *et al.* (incl. **APW**), *The clustering of galaxies in the completed SDSS-III Baryon Oscillation Spectroscopic Survey: cosmological analysis of the DR12 galaxy sample*, MNRAS, 470, 2617, 2017 (arXiv:1607.03155) [3140 citations]

27 **Price-Whelan, A. M.**; Foreman-Mackey, D., *schwimmbad: A uniform interface to parallel processing pools in Python*, JOSS, 2, 357, 2017 [43 citations]

26 Pearson, S.; **Price-Whelan, A. M.**; Johnston, K. V., *Gaps and length asymmetry in the stellar stream Palomar 5 as effects of Galactic bar rotation*, Nature Astronomy, 1, 633, 2017 (arXiv:1703.04627) [143 citations]

25 Johnston, K. V.; **Price-Whelan, A. M.**; Bergemann, M.; Laporte, C. *et al.*, *Disk Heating, Galactoseismology, and the Formation of Stellar Halos*, MDPI: galaxies, 5, 44, 2017 (arXiv:1709.00491) [12 citations]

24 Li, T. S. *et al.* (incl. **APW**), *Exploring Halo Substructure with Giant Stars. XV. Discovery of a*

Connection between the Monoceros Ring and the Triangulum-Andromeda Overdensity?, ApJ, 844, 74, 2017 (arXiv:1703.05384) [37 citations]

23 Oh, S.; **Price-Whelan, A. M.**; Hogg, D. W.; Morton, T. D. *et al.*, *Comoving Stars in Gaia DR1: An Abundance of Very Wide Separation Comoving Pairs*, AJ, 153, 257, 2017 (arXiv:1612.02440) [174 citations]

22 Sesar, B.; Fouesneau, M.; **Price-Whelan, A. M.**; Bailer-Jones, C. A. L. *et al.*, *A Probabilistic Approach to Fitting Period-luminosity Relations and Validating Gaia Parallaxes*, ApJ, 838, 107, 2017 (arXiv:1611.07035) [51 citations]

21 **Price-Whelan, A. M.**; Hogg, D. W.; Foreman-Mackey, D.; Rix, H., *The Joker: A Custom Monte Carlo Sampler for Binary-star and Exoplanet Radial Velocity Data*, ApJ, 837, 20, 2017 (arXiv:1610.07602) [150 citations]

20 Charisi, M.; Bartos, I.; Haiman, Z.; **Price-Whelan, A. M.** *et al.*, *A population of short-period variable quasars from PTF as supermassive black hole binary candidates*, MNRAS, 463, 2145, 2016 (arXiv:1604.01020) [274 citations]

19 **Price-Whelan, A. M.**; Sesar, B.; Johnston, K. V.; Rix, H., *Spending Too Much Time at the Galactic Bar: Chaotic Fanning of the Ophiuchus Stream*, ApJ, 824, 104, 2016 (arXiv:1601.06790) [64 citations]

18 **Price-Whelan, A. M.**; Johnston, K. V.; Valluri, M.; Pearson, S. *et al.*, *Chaotic dispersal of tidal debris*, MNRAS, 455, 1079, 2016 (arXiv:1507.08662) [76 citations]

17 Charisi, M.; Bartos, I.; Haiman, Z.; **Price-Whelan, A. M.** *et al.*, *Multiple periods in the variability of the supermassive black hole binary candidate quasar PG1302-102?*, MNRAS, 454, 2015 (arXiv:1502.03113) [30 citations]

16 **Price-Whelan, A. M.**; Johnston, K. V.; Sheffield, A. A.; Laporte, C. F. P. *et al.*, *A reinterpretation of the Triangulum-Andromeda stellar clouds: a population of halo stars kicked out of the Galactic disc*, MNRAS, 452, 676, 2015 (arXiv:1503.08780) [99 citations]

15 Sesar, B. *et al.* (incl. **APW**), *The Nature and Orbit of the Ophiuchus Stream*, ApJ, 809, 59, 2015 (arXiv:1501.00581) [32 citations]

14 Alam, S. *et al.* (incl. **APW**), *The Eleventh and Twelfth Data Releases of the Sloan Digital Sky Survey: Final Data from SDSS-III*, ApJS, 219, 12, 2015 (arXiv:1501.00963) [2434 citations]

13 Pearson, S.; Küpper, A. H. W.; Johnston, K. V.; **Price-Whelan, A. M.**, *Tidal Stream Morphology as an Indicator of Dark Matter Halo Geometry: The Case of Palomar 5*, ApJ, 799, 28, 2015 (arXiv:1410.3477) [84 citations]

12 **Price-Whelan, A. M.**; Hogg, D. W.; Johnston, K. V.; Hendel, D., *Inferring the Gravitational Potential of the Milky Way with a Few Precisely Measured Stars*, ApJ, 794, 4, 2014 (arXiv:1405.6721) [60 citations]

11 Anderson, L. *et al.* (incl. **APW**), *The clustering of galaxies in the SDSS-III Baryon Oscillation Spectroscopic Survey: baryon acoustic oscillations in the Data Releases 10 and 11 Galaxy samples*, MNRAS, 441, 24, 2014 (arXiv:1312.4877) [1507 citations]

10 Ahn, C. P. *et al.* (incl. **APW**), *The Tenth Data Release of the Sloan Digital Sky Survey: First Spectroscopic Data from the SDSS-III Apache Point Observatory Galactic Evolution Experiment*, ApJS, 211, 17, 2014 (arXiv:1307.7735) [955 citations]

9 **Price-Whelan, A. M.**; Agüeros, M. A.; Fournier, A. P.; Street, R. *et al.*, *Statistical Searches for Microlensing Events in Large, Non-uniformly Sampled Time-Domain Surveys: A Test Using Palomar Transient Factory Data*, ApJ, 781, 35, 2014 (arXiv:1311.3683) [14 citations]

8 Astropy Collaboration *et al.* (incl. **APW**), *Astropy: A community Python package for astronomy*, A&A, 558, 2013 (arXiv:1307.6212) [14787 citations]

7 Dawson, K. S. *et al.* (incl. **APW**), *The Baryon Oscillation Spectroscopic Survey of SDSS-III*, AJ, 145, 10, 2013 (arXiv:1208.0022) [2109 citations]

6 Ahn, C. P. *et al.* (incl. **APW**), *The Ninth Data Release of the Sloan Digital Sky Survey: First Spectroscopic Data from the SDSS-III Baryon Oscillation Spectroscopic Survey*, ApJS, 203, 21, 2012 (arXiv:1207.7137) [1376 citations]

5 Eisenstein, D. J. *et al.* (incl. **APW**), *SDSS-III: Massive Spectroscopic Surveys of the Distant Universe, the Milky Way, and Extra-Solar Planetary Systems*, AJ, 142, 72, 2011 (arXiv:1101.1529) [2070 citations]

4 Aihara, H. *et al.* (incl. **APW**), *Erratum: "The Eighth Data Release of the Sloan Digital Sky Survey: First Data from SDSS-III" <A href="/abs/2011ApJS..193...29A">(2011, ApJS, 193, 29)</A>*, ApJS, 195, 26, 2011 [59 citations]

3 Blanton, M. R. *et al.* (incl. **APW**), *Improved Background Subtraction for the Sloan Digital Sky Survey Images*, AJ, 142, 31, 2011 (arXiv:1105.1960) [558 citations]

2 Aihara, H. *et al.* (incl. **APW**), *The Eighth Data Release of the Sloan Digital Sky Survey: First Data from SDSS-III*, ApJS, 193, 29, 2011 (arXiv:1101.1559) [1368 citations]

1 **Price-Whelan, A. M.**; Hogg, D. W., *What Bandwidth Do I Need for My Image?*, PASP, 122, 207, 2010 (arXiv:0910.2375) [4 citations]

Preprints & other

25 Behmard, A. *et al.* (incl. **APW**), *M Dwarf Singles and Multis Have Dissimilar Metallicities Indicating Different Origins*, 2026 (arXiv:2609.27013)

24 Tavangar, K. *et al.* (incl. **APW**), *Phase-Spirals Across Galactic Disks II: Using large-scale "macro-spirals" in phase-spiral amplitude to derive perturbation times*, 2026 (arXiv:2609.09543)

23 Pearson, S.; Nibauer, J.; Cunningham, E. C.; **Price-Whelan, A. M.** *et al.*, *Extragalactic Stellar Streams in Time-Dependent Cosmological Halos*, 2026 (arXiv:2609.00526)

22 Kiel Holm, J. *et al.* (incl. **APW**), *Evidence for the First Globular Cluster Stellar Stream beyond the Milky Way*, 2026 (arXiv:2608.12254)

21 Dubay, L. O. *et al.* (incl. **APW**), *A Broken Clock Is Right Twice a Day: [Ce/Mg] Is Not a Universal Chemical Clock*, 2026 (arXiv:2608.26261)

20 SDSS Collaboration *et al.* (incl. **APW**), *The Twentieth Data Release of the Sloan Digital Sky Survey: First All-Sky BOSS Spectra, eROSITA-SDSS-V Mapper Coordinated Observations, and a Preview of the Local Volume Mapper*, 2026 (arXiv:2607.26149)

19 Wagg, T.; Breivik, K.; **Price-Whelan, A. M.**; Renzo, M. *et al.*, *Charting the Galactic Underworld I: Comprehensive simulations of the kinematics, rates, and demographics of Milky Way black holes*, 2026 (arXiv:2607.22814)

18 Han, J. J. *et al.* (incl. **APW**), *A Path to an All-Sky Survey with Roman*, 2026 (arXiv:2602.21280)

[2 citations]

17 Sayeed, M. *et al.* (incl. **APW**), *Looking for Companionship: Radial Velocity Follow-Up of Lithium-Rich Giants with ESPRESSO*, 2025 (arXiv:2510.17966)

16 Sagear, S.; Ballard, S.; Daniel, K. J.; **Price-Whelan, A. M.** *et al.*, *The Orbital Eccentricities of Planets in the Kinematic Thin and Thick Galactic Disks*, 2025 (arXiv:2509.23973) [3 citations]

15 Blanton, M. R. *et al.* (incl. **APW**), *The Future of Astronomical Data Infrastructure: Meeting Report*, 2023 (arXiv:2311.04272)

14 Dey, A. *et al.* (incl. **APW**), *RomAndromeda: The Roman Survey of the Andromeda Halo*, 2023 (arXiv:2306.12302) [7 citations]

13 Han, J. J.; Dey, A.; **Price-Whelan, A. M.**; Najita, J. *et al.*, *NANCY: Next-generation All-sky Near-infrared Community surveY*, 2023 (arXiv:2306.11784) [11 citations]

12 Grunblatt, S. K. *et al.* (incl. **APW**), *Roman CCS White Paper: Adding Fields Hosting Globular Clusters To The Galactic Bulge Time Domain Survey*, 2023 (arXiv:2306.10647) [6 citations]

11 Breivik, K. *et al.* (incl. **APW**), *From Data to Software to Science with the Rubin Observatory LSST*, 2022 (arXiv:2208.02781) [17 citations]

10 Bechtol, K. *et al.* (incl. **APW**), *Snowmass2021 Cosmic Frontier White Paper: Dark Matter Physics from Halo Measurements*, 2022 (arXiv:2203.07354) [70 citations]

9 Katz, D. S. *et al.* (incl. **APW**), *Software Sustainability & High Energy Physics*, 2020 (arXiv:2010.05102) [2 citations]

8 Oladosu, A. *et al.* (incl. **APW**), *Meta-Learning for One-Class Classification with Few Examples using Order-Equivariant Network*, 2020 (arXiv:2007.04459) [4 citations]

7 Hogg, D. W.; **Price-Whelan, A. M.**; Leistedt, B., *Data Analysis Recipes: Products of multivariate Gaussians in Bayesian inferences*, 2020 (arXiv:2005.14199) [14 citations]

6 Ness, M. *et al.* (incl. **APW**), *In Pursuit of Galactic Archaeology: Astro2020 Science White Paper*, 2019 (arXiv:1907.05422)

5 Buckley, M. R.; Hogg, D. W.; **Price-Whelan, A. M.**, *Applying Liouville's Theorem to Gaia Data*, 2019 (arXiv:1907.00987) [3 citations]

4 The MSE Science Team *et al.* (incl. **APW**), *The Detailed Science Case for the Maunakea Spectroscopic Explorer, 2019 edition*, 2019 (arXiv:1904.04907) [91 citations]

3 Breivik, K.; **Price-Whelan, A. M.**; D'Orazio, D. J.; Hogg, D. W. *et al.*, *Stellar multiplicity: an interdisciplinary nexus*, 2019 (arXiv:1903.05094) [4 citations]

2 Bergemann, M. *et al.* (incl. **APW**), *Stellar Astrophysics and Exoplanet Science with the Maunakea Spectroscopic Explorer (MSE)*, 2019 (arXiv:1903.03157) [2 citations]

1 **Price-Whelan, A. M.**; Oh, S.; Spergel, D. N., *Spectroscopic confirmation of very-wide stellar binaries and large-separation comoving pairs from Gaia DR1*, 2017 (arXiv:1709.03532) [18 citations]